MIL-C-25781C (USAF)
27 March 1981
SUPERSEDING
MIL-C-25781B (USAF)
21 JANUARY 1964

MILITARY SPECIFICATION

CONVERTER, LIQUID OXYGEN GCU-3/A

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of converter. The converter is designated GCU-3/A. The converter does not have integral warming coils.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-9050	Valves, Oxygen, Pressure Relief, Aircraft
MIL-C-25516	Connectors, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification For
MIL-C-25666	Converters, Liquid Oxygen, Capacitance Type Gaging, General Specification For

STANDARDS

MILITARY

MIL-STD-831	Test Reports, Preparation Of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC: 1660

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurements should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General specification. Unless otherwise specified herein, the converter shall conform to the requirements specified in MIL-C-25666.

3.2 Qualification. Converters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.3 Design and construction. The converter shall conform to figures 1, 2 and 3. The converter shall have an operating pressure of 70 pounds per square inch gage (psig) and a minimum capacity of 10 liters of liquid oxygen measured at 1 atmosphere.

3.3.1 Strength. The filled converter shall be designed to withstand the dynamic loads that are imposed by the following accelerations:

Vertical down	-	14.0 G
Vertical up	-	6.0 G
Lateral	-	6.0 G

3.3.2 Marking of ports. The ports shall be marked as indicated in figure 1.

3.4 Performance.

3.4.1 Final assembly pressure. The final assembly pressure of the converter shall be 95 psig.

3.4.2 Filling time. The time required to fill the converter shall not exceed 5 minutes.

3.4.3 Evaporation loss. The evaporation loss of the essentially full converter shall not exceed 3 pounds.

3.4.4 Buildup time. The time required for the converter to reach its operating pressure immediately after it has been filled shall not exceed 10 minutes.

3.4.5 Capacitance. The capacitance of the sensing element, after it is installed in the empty converter, shall be 123.5 (plus or minus 1.0) picofarad (pF).

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3.4.6 Permanent volumetric expansion. The permanent volumetric expansion of the inner vessel shall not exceed 10 percent of the total volumetric expansion while the inner vessel is subjected to a pneumatic or hydrostatic expansion pressure of 120 psig.

3.4.7 Delivery rate. The converter shall deliver constant flows of 72 ± 10 liters of gaseous oxygen per minute for not less than 1 1/2 hours with the operating pressure (measured on the gaseous phase of the system) remaining between 70 and 80 psig after the first 15 minutes of flow. Pressure in the supply line shall remain between 55 psig and 100 psig during the entire period of flow. Flow at the specified rate shall be initiated within 10 minutes after the converter has reached the operating pressure.

3.4.8 Calibration. The capacitance sensing element shall have characteristics that provide for a linear relationship between electrical capacitance and volume of liquid oxygen sensed. The relationship shall be in accordance with figure 4, using the equations that are specified in 3.4.8.1 to determine tolerances. An example of the use of these equations is as follows: If the actual quantity of liquid oxygen at the 6 liter test point is 6.1 liters, the measured capacitance must be between C minimum and C maximum.

C minimum = 5.7 x 6.1 + 122.3 = 157.1 pF
C maximum = 5.9 x 6.1 + 124.7 = 160.7 pF

3.4.8.1 Equations. The following equations shall be used to calculate the minimum and the maximum capacitances.

Values less than 2 liters	Values greater than 2 liters
C minimum = 5.6 L + 122.5	C minimum = 5.7 L + 122.3
C maximum = 6.0 L + 124.5	C maximum = 5.9 L + 124.7

C = Capacitance in picofarads (pF)
L = Exact quantity in liters of liquid at test point

3.4.9 Operating temperature. The operating pressure shall remain between 70 and 85 psig after the first 15 minutes of flow at the low temperature and shall remain between 65 and 80 psig after the first 15 minutes of flow at the high temperature. An intermittent increase or decrease of pressure is allowable provided these variances do not exceed a 2 minute interval for any one pressure cycle.

3.4.10 Vibration. After the converter has been vibrated in accordance with MIL-C-25666, the evaporation loss of the converter in the buildup condition shall not exceed 3 pounds of oxygen.

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3.5 Pressure relief valves. The pressure relief valves shall conform to type II of MIL-V-9050 and shall be of such size and shape to fit the over-all dimensions specified for the converter.

3.6 Check valve. The check valve used in the system shall be designed to open at 5 ± 2 psig.

3.7 Dimensions. The dimensions of the converter shall conform to figures 1 and 2.

3.8 Weight. The weight of the converter shall not exceed 25 pounds.

3.9 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection required by this specification is classified as follows:

a. Qualification inspection.

b. Quality conformance inspection.

4.3 Qualification inspection. The qualification inspection shall consist of all of the methods of inspection that are described in MIL-C-25666.

4.3.1 Samples. At least two converters shall be subjected to the qualification inspection. One converter shall be submitted to the qualifying activity (see 6.3). The converter that is submitted to the qualifying activity may be a tested converter provided it has not been subjected to the gunfire test that is described in MIL-C-25666.

4.3.2 Report. Three copies of a qualification inspection report

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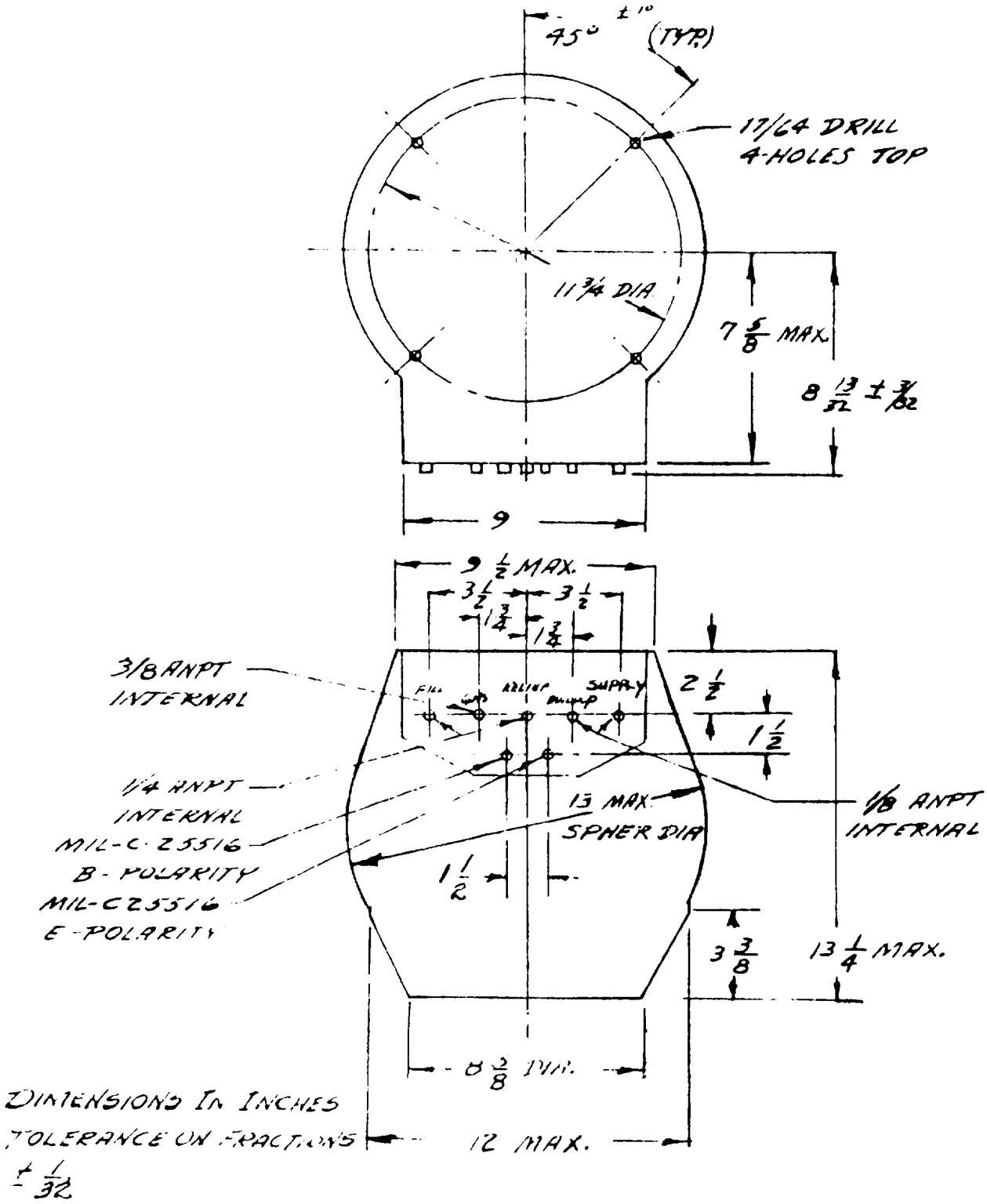


FIGURE 1. 10 Liter converter

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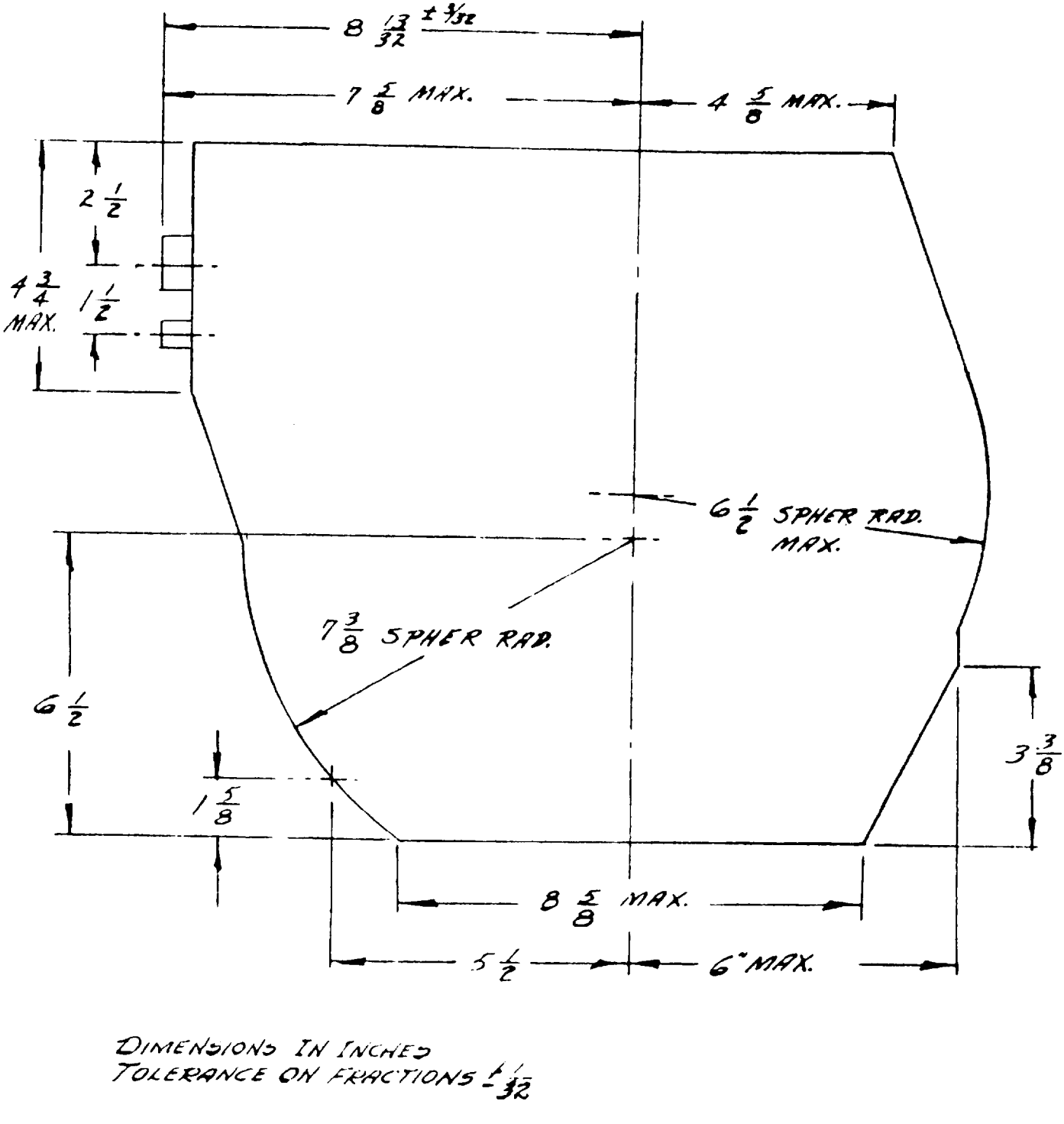


FIGURE 2. 10 Liter converter, side view

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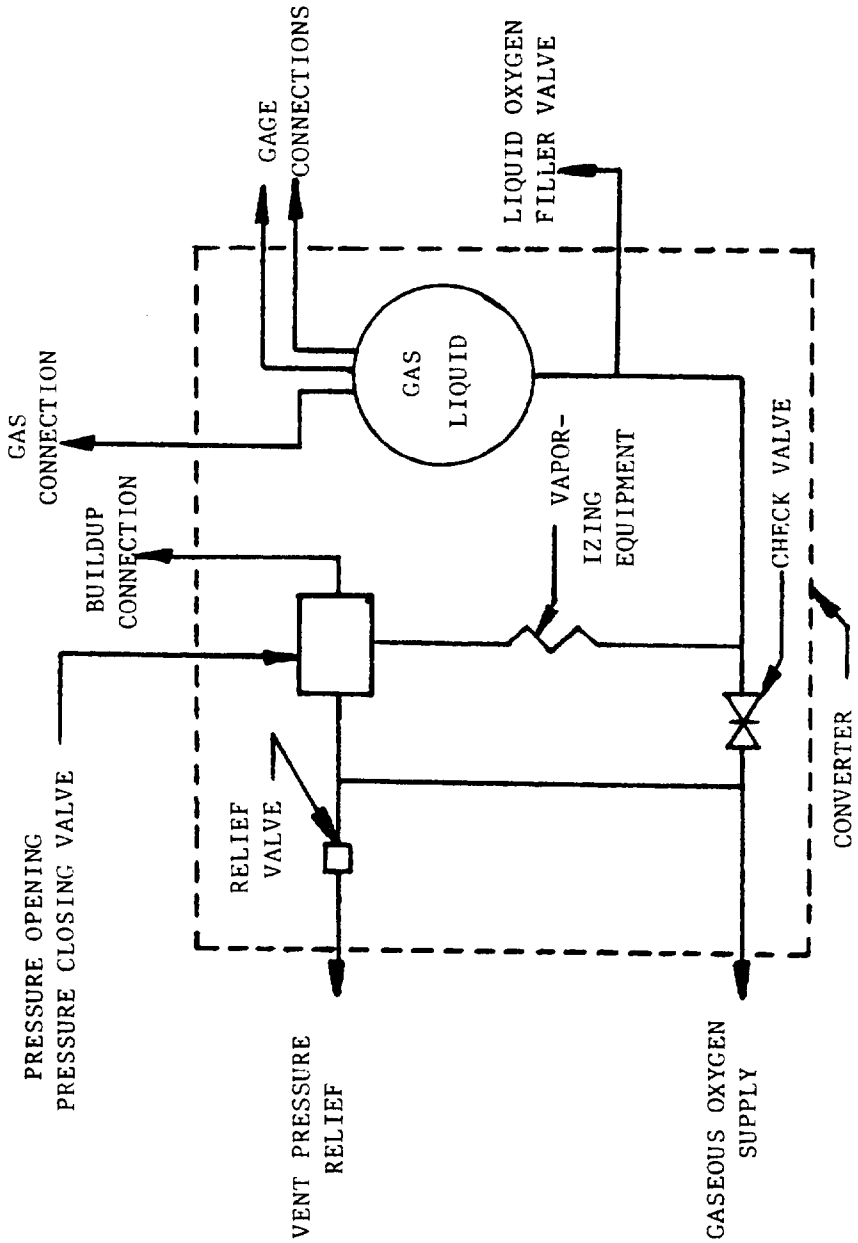
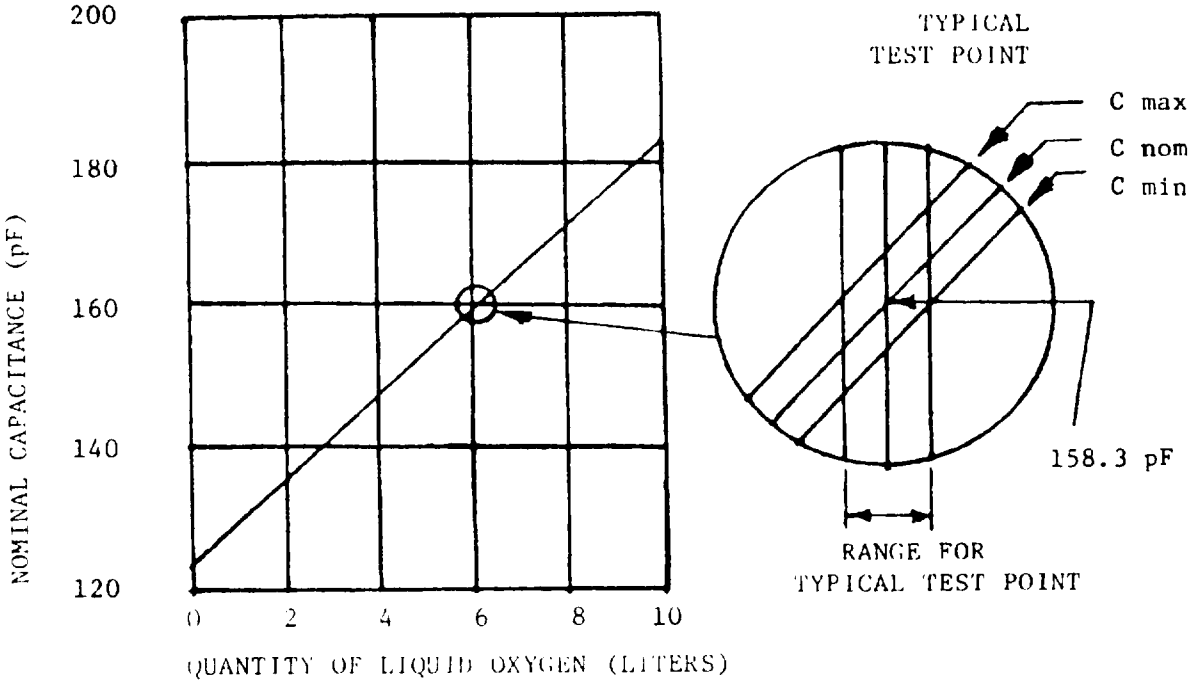


Figure 3. Schematic diagram of 10-liter converter with capacitance gaging element.

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LIQUID OXYGEN (LITERS)	NOMINAL CAPACITANCE (pF)
0	123.5
2	135.1
4	146.7
6	158.3
8	16919
10	181.5

FIGURE 4. Relationship between capacitance and volume of oxygen sensed

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that has been prepared in accordance with MIL-STD-831 shall be submitted to the qualifying activity (see 6.3).

4.4 Quality conformance inspection. The quality conformance inspection shall be in accordance with MIL-C-25666.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with MIL-C-25666 (see 6.2).

6. NOTES

6.1 Intended use. The converter covered by this specification is intended to be used on the F-106A/B aircraft to provide storage and pressurization of 10 liters of liquid oxygen at 70 psig nominal pressure.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation and packaging and packing of MIL-C-25666 (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they proposed to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for this Qualified Products List is HQ AFLC CASO/LODS, Battle Creek MI 49016 and information pertaining to qualification of products may be obtained from that activity.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 99

Reviewer:
Air Force - 11

Preparing Activity:
Air Force - 99

Project Number:
1660-F419

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-25781C

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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